
Decodability Is Not Localization: Auditing Hidden-State Error Probes in Qwen2.5-Math-1.5B-Instruct

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Abstract

1 We audit whether hidden-state probes identify when mathematical reasoning first
2 becomes invalid, or instead exploit easier correlates of an invalid prefix. In a case
3 study of Qwen2.5-Math-1.5B-Instruct on ProcessBench, a linear probe predicts
4 the persistent target *invalid so far* with held-out AUROC 0.866. Equally tuned
5 nuisance baselines collapse most of that margin: prefix TF-IDF reaches 0.760,
6 structural metadata 0.780, and a joint text-plus-metadata model with final-answer
7 correctness reaches 0.874—numerically *above* the hidden state (paired difference
8 -0.008 , 95% CI $[-0.028, 0.013]$). The hidden probe’s remaining value is nar-
9 rower and task-specific: it localizes the annotated onset better (Process F1 0.393
10 versus 0.294; paired difference 0.099 $[0.040, 0.158]$) and rejects correct traces
11 that the joint shortcut model falsely alarms on, while it localizes exactly on only
12 28.9% of erroneous traces and falsely alarms on 38.8% of correct traces. A post-
13 hoc, metadata-matched transition probe obtains AUROC 0.769 on held-out pairs;
14 matched-covariate balance diagnostics and reuse quantification keep it explicitly
15 exploratory. Two verdict readouts fail their behavioral validity checks, so the in-
16 tervention remains a failed pilot. We conclude that decodability, incremental in-
17 formation over tuned nuisances, localization, calibration, and demonstrated causal
18 use are distinct validation levels, and we extract a concrete checklist for when an
19 internal pattern is safe to treat as a candidate discovery.

20 1 Introduction

21 Process supervision asks whether each step of a solution is valid, rather than whether the final answer
22 happens to be correct (Cobbe et al., 2021; Lightman et al., 2023). A related mechanistic question is
23 whether a solving model’s own representation changes when its reasoning first goes wrong. Probe
24 accuracy alone cannot answer it: if every boundary after the first error is labeled positive, a classifier
25 may recognize accumulated inconsistency, solution length, generator style, or final-answer outcome
26 without responding at the onset.

27 The *Interpretability for Discovery* framing makes the stakes explicit: a linearly decodable pattern
28 is a *candidate*, not yet discovered knowledge. Turning a candidate into a claim requires validation
29 at distinct levels, and failing any one changes what the pattern can be used for. This paper is a
30 worked negative example of that validation sequence. We test four claims that are often collapsed:
31 (i) invalid prefixes are linearly decodable; (ii) a score localizes the first error; (iii) the signal transfers
32 across benchmark sources; and (iv) the model behaviorally uses the decoded direction. We test them
33 in one bounded setting: Qwen2.5-Math-1.5B-Instruct on ProcessBench (Yang et al., 2024; Zheng
34 et al., 2025), with problem-grouped splits and a persistent invalid-so-far target, audited by prefix-
35 text, metadata, and joint nuisance controls fit under the same validation budget as the probe, source
36 and generator stratification, an onset-aligned analysis, and a matched transition probe.

Table 1: Closest hidden-state verifier work. “Use” records the reported downstream or causal test, not a judgment of overall paper quality.

Work	Supervision / task	Scope	Onset evaluation	Use test
Sun et al.	arithmetic error	addition, GSM8K	no first-error metric	re-prompting
ReProbe	step credibility	math, planning, QA	step scoring	test-time scaling
STEP	step quality	reasoning traces	early step scoring	pruning
Ours	human first error	four ProcessBench sources	first crossing + transition	failed pilot

The audit changes the interpretation. The hidden-state probe’s pooled AUROC is high, but an equally tuned joint shortcut model combining prefix text, structural metadata, and final-answer correctness exceeds it numerically. What survives is a narrower, task-specific advantage: trace-level localization and correct rejection, both weak in absolute terms, with poor length- and source-dependent calibration and an uninterpretable behavioral assay. We contribute a documented failure mode: strong prefix decoding and cross-source ranking can coexist with no incremental pooled information over tuned nuisances.

2 Related work and contribution boundary

External process reward models learn step quality from human or model supervision (Lightman et al., 2023; Li et al., 2023). ProcessBench annotates the earliest error in 3,400 model-generated traces from GSM8K, MATH, OlympiadBench, and Omni-MATH (Zheng et al., 2025; Hendrycks et al., 2021; He et al., 2024; Gao et al., 2024). We repartition these benchmark sources for representation analysis; they are heterogeneous sources, not independent domains or evidence of model-family generalization.

Closest to this work, Sun et al. probe arithmetic errors and use the probe for selective re-prompting; ReProbe learns hidden-state step scorers for test-time scaling; and STEP uses hidden-state scores to prune reasoning traces (Sun et al., 2025; Ni et al., 2026; Liang et al., 2026). Table 1 states the comparison at the level needed for our claim: our novelty is not that hidden states predict reasoning quality, but the joint audit of a human first-error target using first-crossing, matched-transition, source-transfer, equally tuned nuisance, and failed-assay evidence.

Probing work cautions that decodability need not imply behavioral use (Hewitt and Liang, 2019; Belinkov, 2022; Elazar et al., 2021); an additive intervention is evidence only if its behavioral readout first distinguishes the relevant baseline states.

3 Experimental design

3.1 Data flow, model, and split

Each ProcessBench trace supplies a problem, generator identity, written steps, source, final-answer correctness, and a human first-error index e_i , with $e_i = -1$ for fully correct traces. Traces sharing normalized problem text form a group. Seed 42 assigns groups within source $\times$ error-status strata to 60% train, 20% validation, and 20% test: of 3,400 attempted traces, 3,360 are retained (2,187 with an annotated error), yielding 24,909 written-step boundaries and 669 held-out traces. A 2,048-token limit excludes a complete trace rather than truncating it. Appendix A gives the full attempted/retained accounting by split and source.

We run Qwen2.5-Math-1.5B-Instruct in bfloat16 and record its 1,536-dimensional residual-stream state at each written step and each of 29 hidden-state indices. The completed extraction used batch size 16 on one NVIDIA A100 GPU. Locked package versions, compute, storage, and licenses appear in Appendix D; artifact identity fixes the model, context length, dtype, and dataset SHA-256.

3.2 Persistent target, probe, and trace-level estimands

At boundary k of trace i , the frozen primary target is

$$y_{ik} = \mathbb{I}[e_i \geq 0 \wedge k \geq e_i]. \quad (1)$$

It means *invalid so far*; it does not label only the first invalid step. A class-balanced, L2 logistic probe is fit separately at each index after train-set standardization. Validation AUROC selects $C \in \{0.01, 0.1, 1, 10\}$; validation Process F1 selects a threshold on a 0.005 grid from 0.05 to 0.95 and then the layer. The selected index is 23, $C = 0.01$, threshold 0.645; it is refit on train plus validation and evaluated once on test.

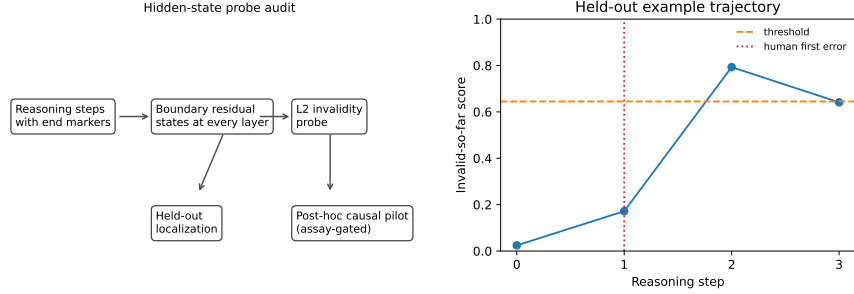


Figure 1: The persistent target and first-crossing evaluation. A causal forward pass produces one boundary state per step. In the example, the score crosses one step after the annotated error.

80 The predicted first error is the first threshold crossing, or -1 if none occurs. We report boundary
 81 AUROC and average precision, exact error-boundary accuracy among erroneous traces, correct-trace
 82 rejection, their harmonic mean (Process F1), complete-trace accuracy, and detection within one or
 83 two steps; the localization metrics weight traces equally. Primary intervals use 1,000 whole-trace
 84 bootstrap draws; the post-hoc paired comparisons use 2,000 draws because they were added in a
 85 later audit configuration, which we state rather than presenting one frozen protocol.

86 3.3 Shortcut, joint-control, and weighting audits

87 Every nuisance control receives the same model-selection budget as the hidden-state probe: L2 reg-
 88 ularization is selected from $\{0.01, 0.1, 1, 10\}$ by validation AUROC, the threshold is selected on the
 89 same validation scores, and the selected model is refit on train plus validation before the single test
 90 evaluation. Each control is also fit with sample weights inverse to boundary counts, so every trace
 91 contributes equal total training loss, mirroring a trace-equal objective rather than reweighting evalua-
 92 tion after fitting. Current-step TF-IDF uses the destination step; prefix TF-IDF uses the problem and
 93 all reasoning text through the current boundary; structural metadata uses source, generator, absolute
 94 and fractional step position, trace length, and token count. Two *joint* baselines combine prefix text,
 95 all structural metadata, and—in the stronger variant—final-answer correctness, so complementarity
 96 across separately fit controls cannot be mistaken for a hidden-state advantage.
 97 Final-answer correctness is a benchmark annotation, not an online signal: it records whether the
 98 trace’s final answer matches the benchmark reference solution, for every trace regardless of where
 99 an error occurs. We use it as a diagnostic upper bound on nuisance predictability, not a deployable
 100 baseline. A shuffled-label hidden probe checks the fitting procedure. We report paired probe-minus-
 101 control intervals from whole-trace resampling, pooled and within final-answer-correctness strata.
 102 The primary objective weights each boundary equally in *evaluation*; the trace-equal refit above
 103 addresses the training asymmetry. Descriptive subgroup outcomes stratify by source, generator,
 104 final-answer correctness, error position, and length and token-count quartiles.

105 3.4 Post-hoc onset analyses

106 The following analyses were designed after the primary test result and are labeled post-hoc. We cir-
 107 cularly shift each frozen score trajectory 5,000 times, preserving its values while breaking alignment
 108 to the annotation. We compare the score jump at each noninitial error with a transition from a fully
 109 correct trace, preferring the same source and generator and then minimizing differences in relative
 110 position, trace length, and token count.

111 The transition probe is a separate analysis of $h_{i,e_i} - h_{i,e_i-1}$. Every eligible erroneous transition is
 112 paired with a correct-trace transition in the same frozen partition, giving 1,152 train, 387 validation,
 113 and 380 test pairs; the pairing reuses correct traces (median 2 pairs per placebo trace, 90th percentile
 114 6, maximum 46, from 687 unique placebos), and matched groups are close on relative position but
 115 retain token-count and length differences (standardized differences 0.28 and 0.15). Layer and C are
 116 selected by validation AUROC; index 21 and $C = 0.01$ are selected. Controls predict from position,
 117 destination-step TF-IDF, or shuffled labels. The protocol is fully specified and executable, but the
 118 test set was not untouched when the analysis was conceived, so the result is exploratory regardless
 119 of its interval.

Table 2: Equally tuned held-out comparison. All rows use the same validation C grid, validation threshold selection, and trace-equal training weights. Paired intervals are hidden minus control; positive Δ favors the hidden state.

Predictor	AUROC	Proc. F1	Exact	Corr. rej.	Δ AUROC [95% CI]
Hidden state	.866	.393	.289	.612	—
Prefix TF-IDF	.760	.281	.293	.477	.106 [.083,.132]
Structural metadata	.780	.147	.164	.194	.086 [.065,.107]
Joint text + metadata	.810	.210	.208	.278	.056 [.037,.073]
Metadata + final outcome	.854	.262	.335	.646	.012 [−.012,.036]
Joint + final outcome	.874	.294	.430	.895	−.008 [−.028,.013]

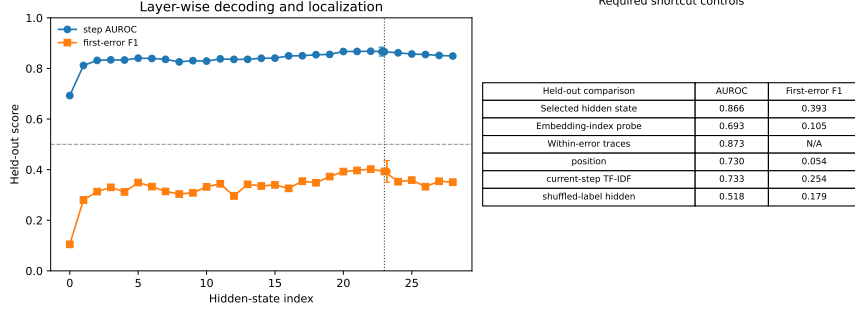


Figure 2: Layer-wise held-out performance and original controls. The embedding-index row is a linear probe at hidden-state index 0, included as a depth comparison rather than a lexical control. Process F1 is N/A within error-only traces because correct-trace rejection is undefined. The equally tuned controls appear in Table 2.

4 Results

4.1 A tuned joint shortcut model matches or exceeds pooled AUROC

The selected probe reaches AUROC 0.866 (95% CI [0.849, 0.884]), average precision 0.843 ([0.813, 0.870]), and step F1 0.743 ([0.716, 0.769]). Within erroneous traces, first-step centering leaves pooled AUROC at 0.881 ([0.862, 0.900]), and mean per-trace AUROC is 0.968 ([0.958, 0.976]). Stable trace offsets therefore do not explain all discrimination.

Table 2 shows the equally tuned comparison. Prefix text reaches AUROC 0.760, structural metadata 0.780, and their joint model 0.810: the controls’ errors are partly complementary. Adding final-answer correctness yields the strongest nuisance model at AUROC 0.874, numerically above the hidden probe’s 0.866 (paired hidden-minus-joint difference -0.008 [−0.028, 0.013], which includes zero). Under a maximally fair tuning budget, the hidden state shows *no incremental pooled-ranking information* over the best nuisance predictor.

The decomposition matters more than the pooled number. The hidden probe retains a Process F1 advantage of 0.099 ([0.040, 0.158]) and an exact-localization advantage of 0.113 ([0.064, 0.163]), because the joint model crosses threshold on most correct traces (correct rejection 0.612 versus 0.895; hidden-minus-joint difference -0.283 [−0.347, −0.218]). What the hidden state contributes is temporal specificity—when to start alarming—not better separation of invalid prefixes in aggregate. Within final-answer-correctness strata, the hidden AUROC advantage is confined to traces whose final answer is correct (0.073 [0.039, 0.107]) and vanishes among incorrect-outcome traces (-0.001 [−0.022, 0.021]), consistent with the outcome field carrying most pooled information when it is informative.

Equal trace weighting at evaluation changes AUROC by +0.001 (95% paired CI [−0.007, 0.010]). Complete-trace accuracy falls from 0.491 in the shortest length quartile to 0.264 in the longest, with heterogeneous source-level estimates (Appendix B).

4.2 First-error localization remains weak

The hidden score localizes 28.9% of erroneous traces exactly, 59.3% within one step, and 72.7% within two. It crosses somewhere on 88.7% of erroneous traces, but also on 38.8% of fully correct

traces, and among detections it is late by +0.59 steps on average ([+0.36, +0.80]). Process F1 is 0.393 ([0.351, 0.436]); complete-trace accuracy is 0.404. Post-hoc temporal tests show more alignment than these threshold metrics alone suggest. Exact localization is 0.289 versus 0.164 under circular shifts ($p = 0.0002$). Across 380 noninitial errors, the mean onset jump is 0.257; matched correct transitions rise 0.113, for a paired difference of 0.144 ([0.096, 0.193]). The score continues from 0.341 immediately before the error to 0.559 at onset, 0.680 one step later, and 0.731 two steps later: a gradual transition into a persistent state, not a sharp alarm. The transition task excludes 52 test traces whose first error occurs at step 0, which may differ systematically from the 380 eligible ones. The matched transition probe reaches AUROC 0.769 ([0.713, 0.820]), average precision 0.741 ([0.672, 0.816]), and paired accuracy 0.753 ([0.676, 0.824]); position, destination-step TF-IDF, and shuffled-label controls reach 0.63, 0.67, and 0.58. Given the placebo reuse and retained covariate imbalance documented above, this supports decodability of an onset-related change on the already-inspected split; sensitivity refits under one-to-one and inverse-reuse matching, and replication on untouched data, are needed before any confirmatory reading.

4.3 Source ranking transfers; model generalization is untested

Separate probes trained on one ProcessBench source and tested on another have mean off-diagonal AUROC 0.805 versus 0.845 on the diagonal; every off-diagonal cell exceeds 0.739. Thresholded Process F1 is less stable: 0.261 off diagonal versus 0.336 on diagonal, with individual off-diagonal values from 0.055 to 0.371. These cells show cross-source ranking within one benchmark construction, not transfer to another model size, family, or dataset.

4.4 Scope of the evidence

This paper is an existence proof of a failure mode in one small math-tuned model on one benchmark construction—decodability without incremental pooled information, localization, calibration, or causal validation—not a measurement of its prevalence across models. A second-model replication under the frozen protocols reported here is deliberately left as follow-up work.

5 Limitations, responsible use, and conclusion

This is a one-model, one-benchmark case study; the title and claims name Qwen2.5-Math-1.5B-Instruct. All added analyses—transition probe, temporal randomization, shortcut and joint controls, trace-equal refits, subgroups—are post-hoc, and all intervals are conditional on one frozen split and the completed selection pipeline: they capture resampling uncertainty, not split or selection sensitivity. No dated preregistration exists. Corrected/error twins lack independent verification, so no patching result is reported; a stronger follow-up would verify twins, reserve an untouched onset set, and validate the readout first.

The detector is unsuitable for grading, certification, or unattended feedback: its 38.8% false-alarm rate on correct traces could penalize valid work, its late detections could let an invalid derivation propagate, and its sensitivity to source, outcome, and length makes distribution shift especially risky. Positive uses are limited to research diagnostics with human or formal verification.

The central empirical finding is a gap with a precise shape: in Qwen2.5-Math-1.5B-Instruct, invalid-so-far prefixes are linearly decodable, but an equally tuned joint text-plus-metadata model with final-answer correctness matches or exceeds the hidden state’s pooled AUROC; what remains is a localization and correct-rejection advantage, itself weak in absolute terms.

For discovery practice, this audit reduces to a checklist. Before treating a decodable internal pattern as discovered knowledge, test and report: (1) whether equally tuned nuisance models—fit jointly, under the same selection budget—recover the pooled performance; (2) whether any residual advantage survives on the task-specific metrics that matter, with paired intervals; (3) whether temporal correspondence holds against matched and shifted controls; (4) whether calibration and subgroup behavior survive the shifts the use case implies; and (5) whether a behavioral readout distinguishes the relevant states before any causal claim. Our candidate passed (1) only in the negative and failed the rest in absolute terms.

References

- Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022. doi: 10.1162/coli_a_00422.

- 200 Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser,
201 Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John
202 Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*,
203 2021. URL <https://arxiv.org/abs/2110.14168>.
- 204 Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. Amnesic probing: Behavioral
205 explanation with amnesic counterfactuals. *Transactions of the Association for Computational*
206 *Linguistics*, 9:160–175, 2021. doi: 10.1162/tac1_a_00359. URL [https://aclanthology.org/](https://aclanthology.org/2021.tac1-1.10/)
207 [2021.tac1-1.10/](https://aclanthology.org/2021.tac1-1.10/).
- 208 Bofei Gao, Feifan Song, Zhe Yang, Zefan Cai, Yibo Miao, Qingxiu Dong, Lei Li, Chenghao Ma,
209 Liang Chen, Runxin Xu, Zhengyang Tang, Benyou Wang, Daoguang Zan, Shanghaoran Quan,
210 Ge Zhang, Lei Sha, Yichang Zhang, Xuancheng Ren, Tianyu Liu, and Baobao Chang. Omni-
211 MATH: A universal olympiad-level mathematics benchmark for large language models. *arXiv*
212 *preprint arXiv:2410.07985*, 2024. URL <https://arxiv.org/abs/2410.07985>.
- 213 Chaoqun He, Renjie Luo, Yuzhuo Bai, Shengding Hu, Zhen Leng Thai, Junhao Shen, Jinyi Hu,
214 Xu Han, Yujie Huang, Yuxiang Zhang, Jie Liu, Lei Qi, Zhiyuan Liu, and Maosong Sun. Olympiad-
215 Bench: A challenging benchmark for promoting AGI with olympiad-level bilingual multimodal
216 scientific problems. *arXiv preprint arXiv:2402.14008*, 2024. URL [https://arxiv.org/abs/](https://arxiv.org/abs/2402.14008)
217 [2402.14008](https://arxiv.org/abs/2402.14008).
- 218 Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song,
219 and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. *arXiv*
220 *preprint arXiv:2103.03874*, 2021. URL <https://arxiv.org/abs/2103.03874>.
- 221 John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Pro-*
222 *ceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and*
223 *the 9th International Joint Conference on Natural Language Processing*, pages 2733–2743. As-
224 sociation for Computational Linguistics, 2019. doi: 10.18653/v1/D19-1275. URL [https://](https://aclanthology.org/D19-1275/)
225 aclanthology.org/D19-1275/.
- 226 Yifei Li, Zeqi Lin, Shizhuo Zhang, Qiang Fu, Bei Chen, Jian-Guang Lou, and Weizhu Chen.
227 Making language models better reasoners with step-aware verifier. In *Proceedings of the 61st*
228 *Annual Meeting of the Association for Computational Linguistics*, pages 5315–5333. Associa-
229 tion for Computational Linguistics, 2023. doi: 10.18653/v1/2023.acl-long.291. URL [https://](https://aclanthology.org/2023.acl-long.291/)
230 aclanthology.org/2023.acl-long.291/.
- 231 Zhixiang Liang, Beichen Huang, Zheng Wang, and Minjia Zhang. Hidden states as early sig-
232 nals: Step-level trace evaluation and pruning for efficient test-time scaling. In *Findings of*
233 *the Association for Computational Linguistics: ACL 2026*, pages 26800–26813. Association
234 for Computational Linguistics, 2026. doi: 10.18653/v1/2026.findings-acl.1336. URL [https://](https://aclanthology.org/2026.findings-acl.1336/)
235 aclanthology.org/2026.findings-acl.1336/.
- 236 Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan
237 Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. *arXiv preprint*
238 *arXiv:2305.20050*, 2023. URL <https://arxiv.org/abs/2305.20050>.
- 239 Jingwei Ni, Ekaterina Fadeeva, Tianyi Wu, Mubashara Akhtar, Jiaheng Zhang, Elliott Ash, Markus
240 Leippold, Timothy Baldwin, See-Kiong Ng, Artem Shelmanov, and Mrinmaya Sachan. ReProbe:
241 Efficient test-time scaling of multi-step reasoning by probing internal states of large language
242 models. In *Proceedings of the 64th Annual Meeting of the Association for Computational Lin-*
243 *guistics*, pages 11667–11689. Association for Computational Linguistics, 2026. doi: 10.18653/
244 [v1/2026.acl-long.536](https://aclanthology.org/2026.acl-long.536/). URL <https://aclanthology.org/2026.acl-long.536/>.
- 245 Yucheng Sun, Alessandro Stolfo, and Mrinmaya Sachan. Probing for arithmetic errors in language
246 models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language*
247 *Processing*, pages 8111–8128. Association for Computational Linguistics, 2025. doi: 10.18653/
248 [v1/2025.emnlp-main.411](https://aclanthology.org/2025.emnlp-main.411/). URL <https://aclanthology.org/2025.emnlp-main.411/>.
- 249 An Yang, Beichen Zhang, Binyuan Hui, Bofei Gao, Bowen Yu, Chengpeng Li, Dayiheng Liu,
250 Jianhong Tu, Jingren Zhou, Junyang Lin, Keming Lu, Mingfeng Xue, Runji Lin, Tianyu Liu,
251 Xingzhang Ren, and Zhenru Zhang. Qwen2.5-Math technical report: Toward mathematical
252 expert model via self-improvement. *arXiv preprint arXiv:2409.12122*, 2024. URL [https://](https://arxiv.org/abs/2409.12122)
253 arxiv.org/abs/2409.12122.
- 254 Chujie Zheng, Zhenru Zhang, Beichen Zhang, Runji Lin, Keming Lu, Bowen Yu, Dayiheng
255 Liu, Jingren Zhou, and Junyang Lin. ProcessBench: Identifying process errors in mathemat-

ical reasoning. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*, pages 987–1011. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.acl-long.50. URL <https://aclanthology.org/2025.acl-long.50/>.

A Additional data accounting

Table 3: Data flow. Attempted / retained traces / exclusions by source and split; boundary counts are written-step boundaries.

Split	Attempted	Retained	Excluded	Boundaries	Error traces	Correct traces
Train	2,041	2,014	27	14,944	1,311	703
Validation	681	677	4	4,980	444	233
Test	678	669	9	4,985	432	237
All	3,400	3,360	40	24,909	2,187	1,173

Table 4: Attempted / retained traces / exclusions by source and split.

Split	GSM8K	MATH	OlympiadBench	Omni-MATH
Train	236/236/0	597/592/5	602/594/8	606/592/14
Validation	80/80/0	198/198/0	204/202/2	199/197/2
Test	84/84/0	205/204/1	194/188/6	195/193/2

The retained boundary counts by split and source are, respectively: train 1,229 / 3,718 / 5,129 / 4,868; validation 414 / 1,303 / 1,708 / 1,555; and test 439 / 1,387 / 1,661 / 1,498 in the source order of the table. The original attempted total was 25,697 boundaries; 24,909 remain after the 40 complete-trace exclusions.

B Descriptive subgroup audit

Table 5: Trace-macro outcomes by ProcessBench source (95% whole-trace bootstrap intervals).

Source	Complete accuracy	Exact error	Correct rejection
GSM8K	.512 [.405,.619]	.238 [.116,.375]	.786 [.658,.902]
MATH	.500 [.426,.569]	.347 [.264,.435]	.723 [.620,.818]
OlympiadBench	.298 [.234,.362]	.207 [.138,.282]	.463 [.340,.585]
Omni-MATH	.358 [.290,.425]	.324 [.253,.401]	.467 [.319,.614]

Among traces with an incorrect final answer, complete accuracy is 0.292 ([0.245, 0.342]). Among traces with a correct final answer it is 0.506 ([0.451, 0.554]); this group mixes valid traces and traces whose reasoning contains an annotated error. Across generators with at least 20 held-out traces, complete accuracy ranges from 0.316 to 0.621. These wide, overlapping estimates are descriptive evidence of heterogeneity rather than generator rankings.

C The failed behavioral pilot

The original CORRECT/INCORRECT readout has baseline AUROC 0.283 and specificity zero. A revised question with conditional Yes/No probabilities remains below chance (AUROC 0.342, specificity 0.203 on 256 balanced boundaries). The intervention code requires both AUROC > 0.5 and nonzero specificity before a learned-direction effect is interpreted. Neither readout passes. Random orthogonal directions cannot substitute for a positive control of assay sensitivity, so the dose response is not causal evidence for or against behavioral use. We report the pilot only to delimit the evidence: no directional intervention was interpreted, so the causal section of the audit is empty by design rather than by null result.

D Reproduction record

The dataset SHA-256 is 447f0a4b35c5747a9f9a3dab1e70d43f71efd501497b8cec668b71337099784a. All stochastic analysis uses seed 42. Exact resolved YAML files and artifact identity files are included with the anonymous supplement. The principal commands are:

```
uv sync --extra dev
uv run math-error-tracing \
  --config configs/experiment1.yaml validate-config
uv run math-error-tracing \
  --config configs/experiment1.yaml run-all
uv run math-error-cpu-followup \
  --config configs/experiment2.yaml
uv run math-error-extended-followup \
  --config configs/experiment3.yaml \
  fit-transition-probe
uv run math-error-extended-followup \
  --config configs/experiment3.yaml \
  analyze-transition-matching
uv run pytest
uv run ruff check src tests
```

Table 6: Compute and licensing record.

Extraction and interventions	one NVIDIA A100 (40 GB), bfloat16, batch 16; single passes over 3,400 traces; ~hours of GPU time
Probe fitting and audits	CPU only; the largest stage bootstraps 2,000 trace resamples
Storage	~1 GB compact predictions plus shards for raw activations
Locked versions	Python 3.10+; PyTorch 2.13.0; Transformers 4.57.6; scikit-learn 1.7.2; NumPy 2.2–2.5; pandas 2.3.3; SciPy 1.15–1.18; datasets 4.8.5
Licenses	ProcessBench: Apache 2.0. Qwen2.5-Math-1.5B-Instruct: Apache 2.0. Analysis code: repository license. Benchmark sources retain their original terms.

Activation extraction and the original intervention used one A100 GPU; probe fitting and all audit analyses are CPU-capable. Extraction is sharded and resumable. The code rejects mismatched dataset, model, dtype, and context-length identities and stores all selected thresholds and predictions. The supplement records the exact environment of each analysis stage in its machine-readable summary files.

NeurIPS Paper Checklist

1. Claims

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